

ATs for Sight

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

<https://cmu-intro-to-accessibility.github.io>

Andrew Begel

Administrivia

- HW4 was due Wednesday 11am.
- Will provide feedback on P2 by this evening.

Disability Pride Pittsburgh!



- Saturday, Sept 26, 12-4pm Schenley Plaza
- Receive one bonus participation point for posting a selfie at Disability Pride to **#in-class** channel on Slack.

Early Course Feedback

- Strongest features of the course
 - Course content
 - In-class activities
 - Hearing our activity answers read out loud
 - Good examples in class
 - We get to try the assistive tech

Early Course Feedback

- Improving Learning - Homework
 - Confusing, need more clear instructions
 - Don't know how to use the tech required
 - Too long
 - Stressful
 - Want the rubrics in advance
- Way too much work when homework and project assignments overlap

Early Course Feedback

- How has the environment helped you feel included?
 - Friendly and diverse student body
 - In-class activities
 - Receive meaningful and respectful responses when I speak
 - Can participate without having to speak out loud
 - Lectures slides posted online
 - Clear topic schedule
 - In-class captioning

Early Course Feedback

- What changes should be made?

- **Fix the homeworks!**

- Go over homework in class.
- Slow down the pace and reduce the workload.

Early Course Feedback

- What are you excited to learn about?
 - Domains: Education, Policy research
 - Disabilities: Color blindness, Intersectional disabilities
 - ATs: Mobility, mental health, physical/hardware, AI, VR/AR, Robotics, communication support
 - How ATs are used and *should* be used in real life

Assistive Technologies for Sight

Glasses

According to glasses.com, the first pair of glasses as we know them for correcting vision were invented sometime around the 13th century in Italy.



Text to Speech

Uses synthesized voices to generate audio or speech output of text on the screen

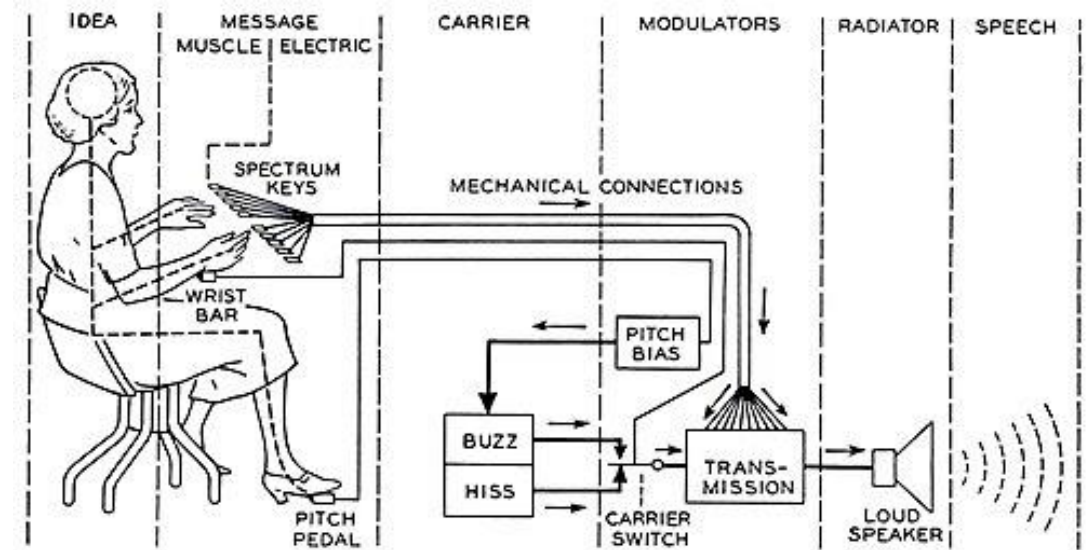


Fig. 8—Schematic circuit of the voder.

Screen Readers

Speaks the words on the screen

Keystroke to move to next area

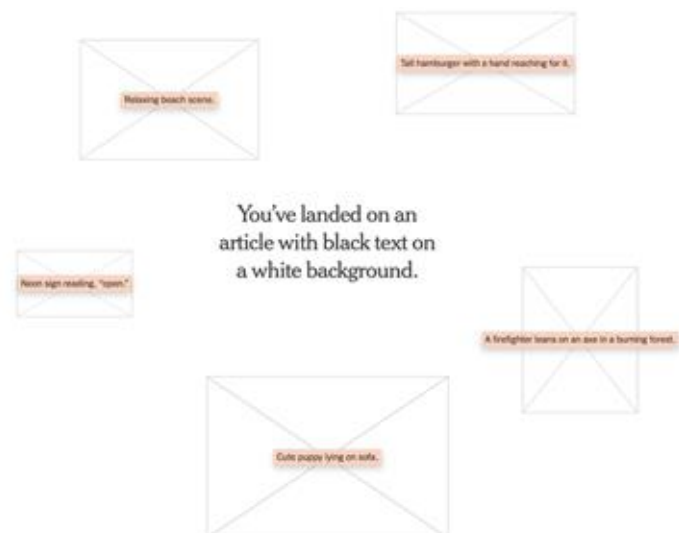
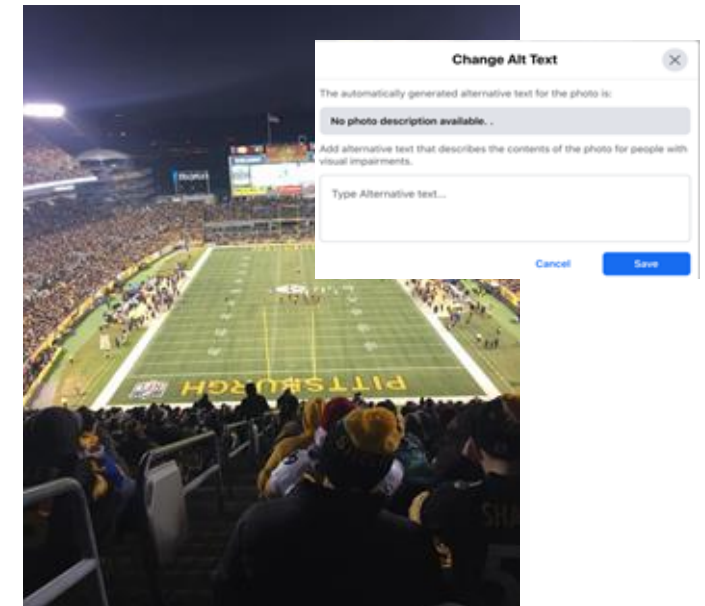
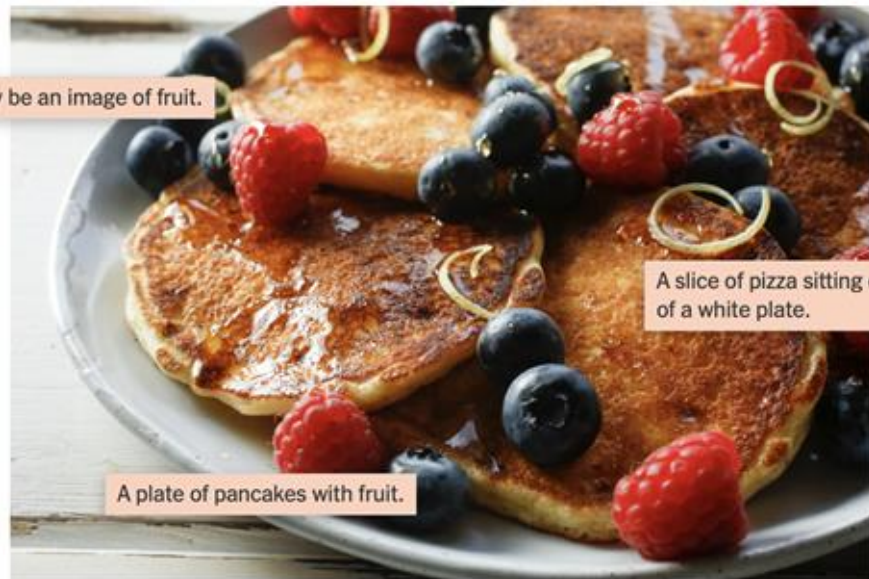
Blind people can operate them amazingly quickly

Examples of Screen Readers:

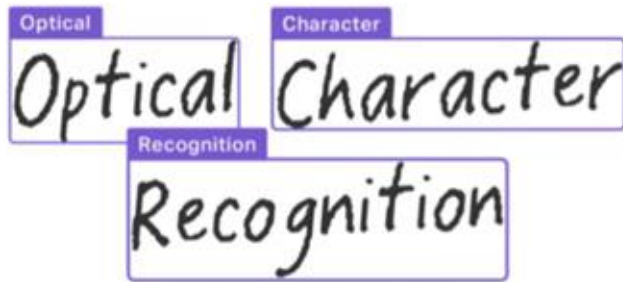
JAWS, NVDA, VoiceOver, TalkBack

Alternative Text

Alternative text provides a way to add text descriptions to image elements



Optical Character Recognition (OCR)



Extracts information from handwritten or printed text.

OCR software can automatically extract data from printed documents, saving money and time for processing them.

Screen Magnification

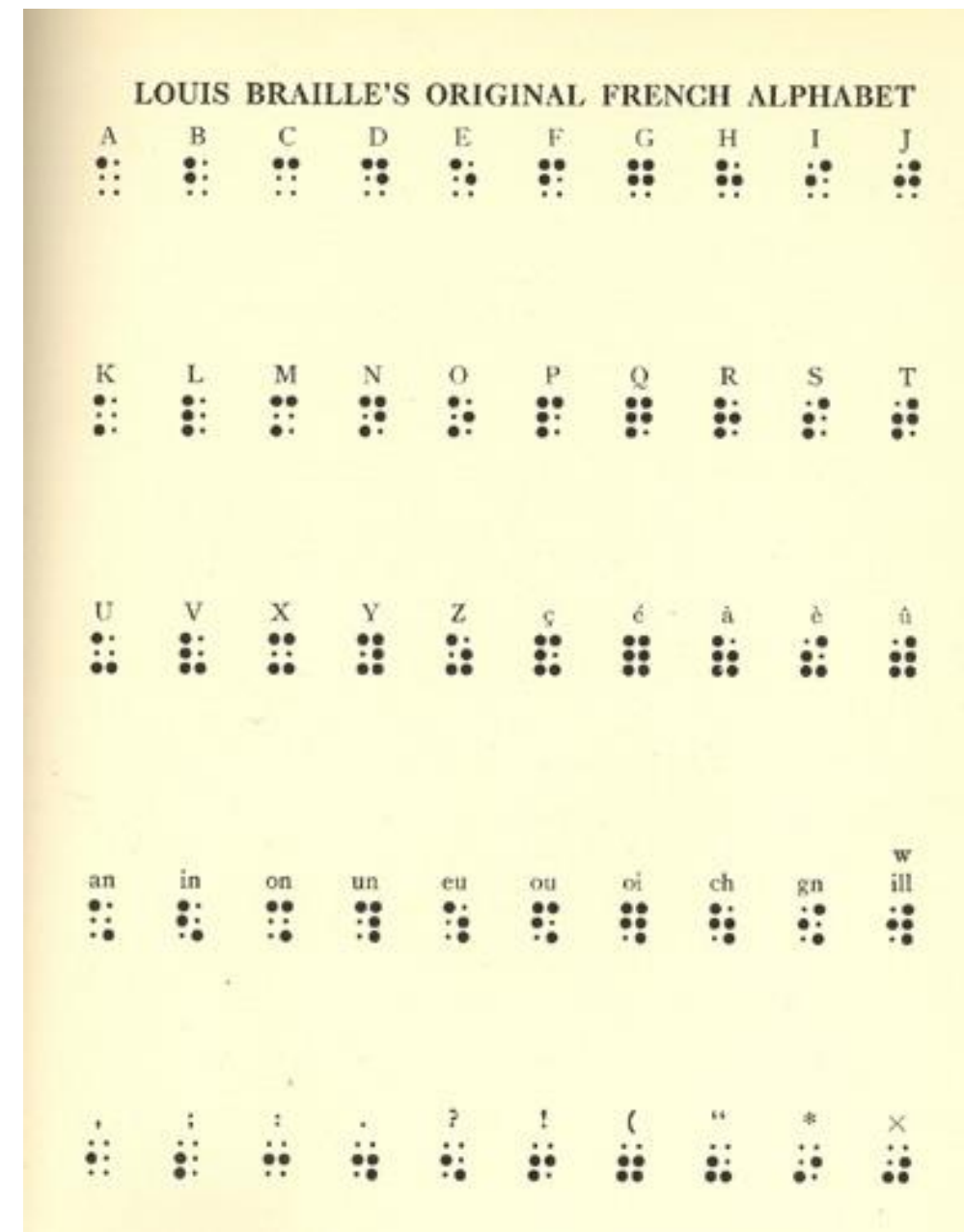
Screen magnification apps are used by individuals with low vision. These apps take a computer or mobile device's visual output and enlarge it to allow for easier and more detailed viewing.

Braille

Braille Code invented by Louis Braille in the 1800s. French Alphabet.

Uniform Braille code not adopted in English-Speaking countries until 1932

Embossed dots. 6 or 8 dot cells



Refreshable Braille Displays

Braille displays provide access to information on a computer screen by electronically raising and lowering different combinations of pins in braille cells

Many have about 40 cells.



Braille Note Takers

More Similar to a Personal Digital Assistant

Read/write files, emails, music, basic web browsing



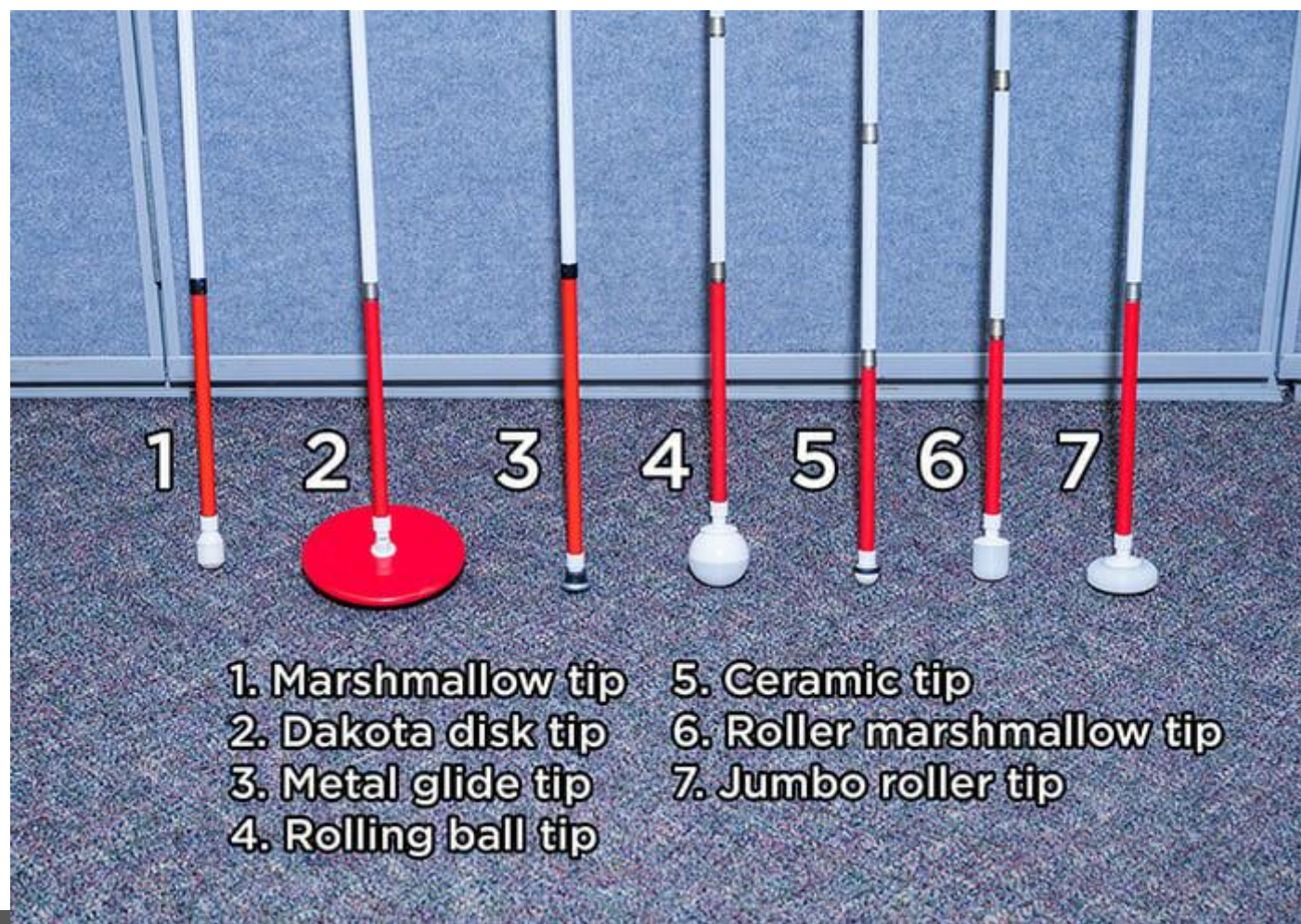
Humanware BrailleNote Touch

White Cane

- People have been using sticks to aid walking for thousands of years.
- *White* cane invented in the early 1900s.
- Lightweight.



White Cane - Tips



White Cane - Tips



Cane – Two Senses, Many Functions

- Orientation and Mobility
 - What is the quality of the ground?
 - What is around me?
 - How do I get through the doorway?
 - How do I cross the street or navigate a sidewalk?
- Sound and Feel
 - What do different surfaces feel like?
 - What do different taps sound like?

How to use a white cane



CaneTroller

Enabling People with Visual Impairments to Navigate Virtual Reality with a Haptic and Auditory Cane Simulation

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Figure 1. (A) A blind user wearing the gear for our VR evaluation, including a VR headset and CaneTroller, our haptic VR controller. (B) The mechanical elements of CaneTroller. (C) Overlays of the virtual scene atop the real scene show how the virtual cane extends past the tip of the CaneTroller device and can interact with the virtual trash bin. (D) The use of CaneTroller to navigate a virtual street crossing: the inset shows the physical environment, while the rendered image shows the corresponding virtual scene. Note that users did not have any visual feedback when using our VR system. The renderings are shown here for clarity.

ABSTRACT

Traditional virtual reality (VR) mainly focuses on visual feedback, which is not accessible for people with visual impairments. We created CaneTroller, a haptic cane controller that simulates white cane interactions, enabling people with visual impairments to navigate a virtual environment by transferring their cane skills into the virtual world. CaneTroller provides three types of feedback: (1) physical resistance generated by a wearable programmable brake mechanism that physically impedes the controller when the virtual cane comes in contact with a virtual object; (2) vibrotactile feedback that simulates the vibrations when a cane hits an object or touches and drags across various surfaces; and (3) spatial 3D auditory feedback simulating the sound of real-world cane interactions. We designed indoor and outdoor VR scenes to evaluate the effectiveness of our controller. Our study showed that CaneTroller was a promising tool that enabled visually impaired participants to navigate different virtual spaces. We discuss potential applications supported by CaneTroller ranging from entertainment to mobility training.

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Author Keywords

Virtual reality; white cane; blindness; visual impairments; haptic feedback; auditory feedback; mobility.

ACM Classification Keywords

H.5.1. Information interfaces and presentation: Multimedia Information Systems. K.4.2. Computers and Society: Assistive technologies for persons with disabilities.

INTRODUCTION

Virtual reality (VR) technologies are maturing quickly and have been widely applied to different fields by researchers and designers, such as entertainment [6,8], education [32,54], and social activities [82]. Researchers have recognized VR's potential in accessibility and developed various VR systems for training and rehabilitation for people with different disabilities [25], such as dyslexia [17,21], stroke [14,33], and ADHD [1,34].

While presenting great potential in a variety of use cases, current VR solutions rely mainly on realistic visual feedback to provide an immersive experience to sighted people, for whom the visual sense is dominant [38]. Most VR applications are not accessible for people with visual impairments (VIPs), preventing them from benefiting from this important class of emerging technologies [28].

Some prior research has explored VR experiences for VIPs. However, they either generated pure auditory virtual reality that has limited capability in describing the shape or specific layout of a virtual space [28,36], or provided

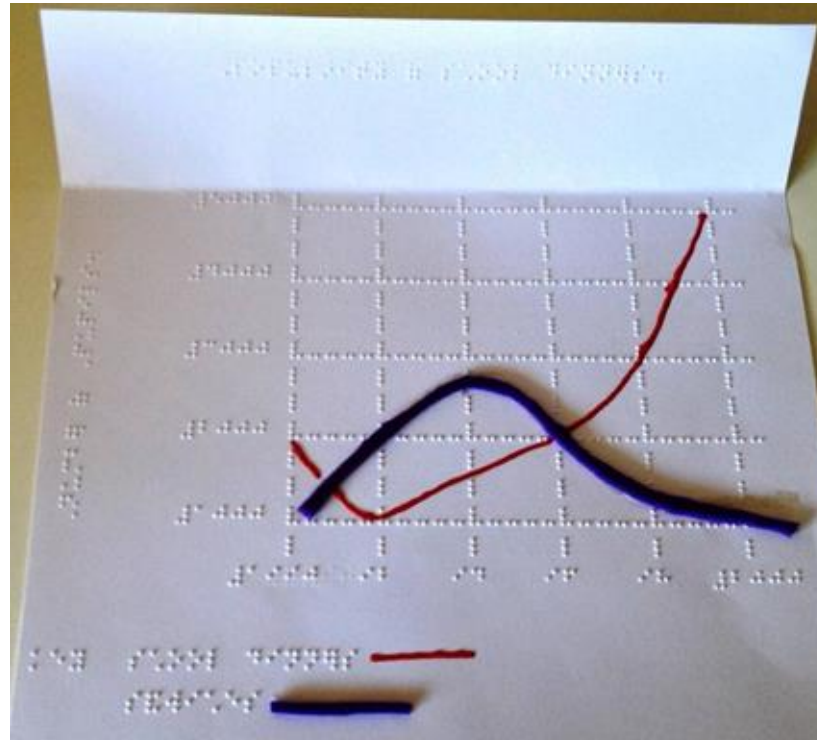


Guide Dogs / Seeing Eye Dogs

- 150,000 years ago, we domesticated dogs
- First training of dogs to assist blind people late 1700s
- 1920-1930s we see guide dog schools in the UK



Tactile Signs and Graphics



GPS Navigation Tools

- 1980s military applications
- ~1996 GPS becomes available for civilians
- Connecting to mapping software and applications



VizWiz

VizWiz

Introduction VizWiz Crowd Agents Scribe

VizWiz



Bigham et al. Nearly Real-Time Answers to Visual Questions. UIST 2010.

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Jeffrey P. Bigham

